Phys 153: Fundamentals of Physics III

Unit #3 – Introduction to Relativity

Aaron Wirthwein

Spring 2027

Introduction to Relativity

Relativity is about how observations made by different observers in different frames of reference are related to one another.

In some sense, it should be obvious that certain aspects of motion depend on the frame of reference; they are frame dependent.

What's not so obvious is that some quantities and relationships appear to be frame independent. Everyone will agree on them regardless of how they are moving (to some extent).

Why Study Relativity?

- Engineers build things that move fast, need to sync precisely, and span large distances. Newtonian mechanics simply doesn't work for these purposes!
- Electromagnetism is already a relativistic theory. Without special relativity, you can't *fully understand* E&M or any technology that relies on it.
- GPS satellites carry atomic clocks that require relativistic calculations to accurately pinpoint locations on earth.
- Forces you to adopt a "first-principles" approach to reasoning because many predictions, conclusions, and observations are difficult to intuitively understand or reconcile on their own.

Conceptual Question

An object is dropped from an airplane flying at constant speed in a straight line at a constant altitude. If there is no air resistance, the falling object will (as seen from the ground)

(A) Lag behind the airplane, but still move forward.

(B) Lag behind the airplane, and fall straight down.

(C) Lag behind the airplane, and move backward.

(D) Remain directly below the airplane.

Principle of Relativity

We postulate that the laws of physics are the same in all inertial frames of reference, in every direction of space, and at all locations and times.

- In Newtonian mechanics, we make the additional assumption that time is universal; simultaneity is absolute.
- In relativistic mechanics, we postulate that the speed of light in vacuum is constant in all inertial frames, independent of the motion of the source or observer.
- As it turns out, absolute simultaneity and a universal light speed are incompatible; one must be false!

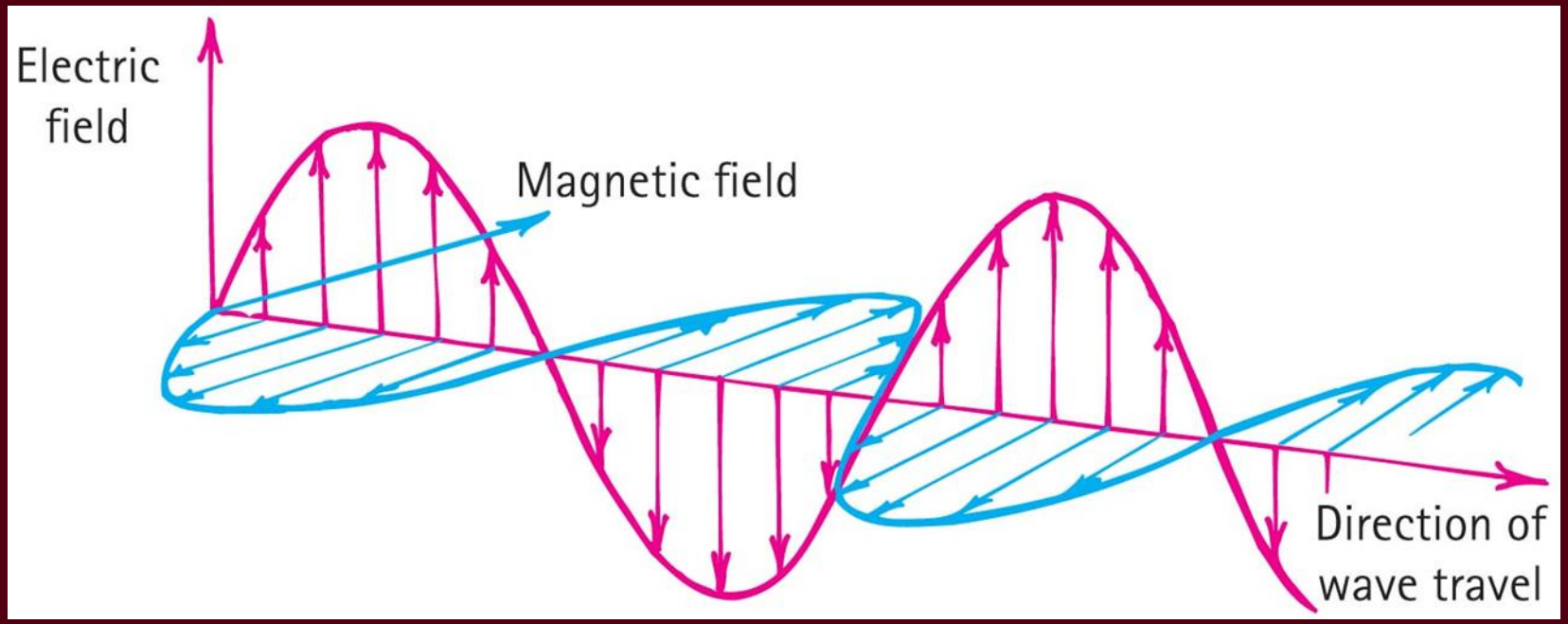
Inertial Frames of Reference

An inertial frame of reference is defined to be one in which Newton's first law is valid. A free particle moves with constant speed in a straight line:

$$\frac{d^2\vec{\mathbf{r}}}{dt^2} = 0 \qquad (1)$$

- From one inertial frame, we can obtain others by performing transformations on the coordinates
  - Rotations
  - Translations
  - Boosts
- All transformations must be linear to preserve Eq. (1)

Can you surf an EM wave?



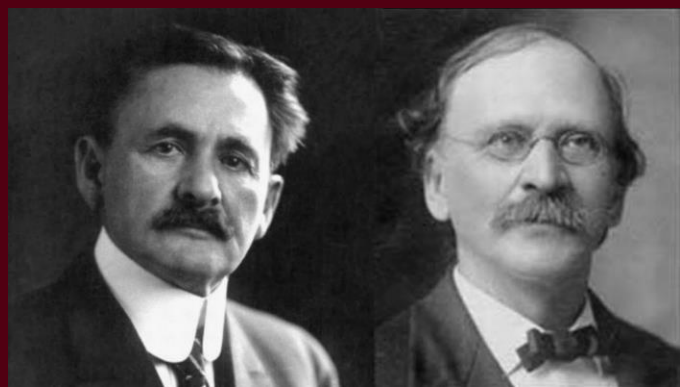
Changing magnetic field

Changing electric field

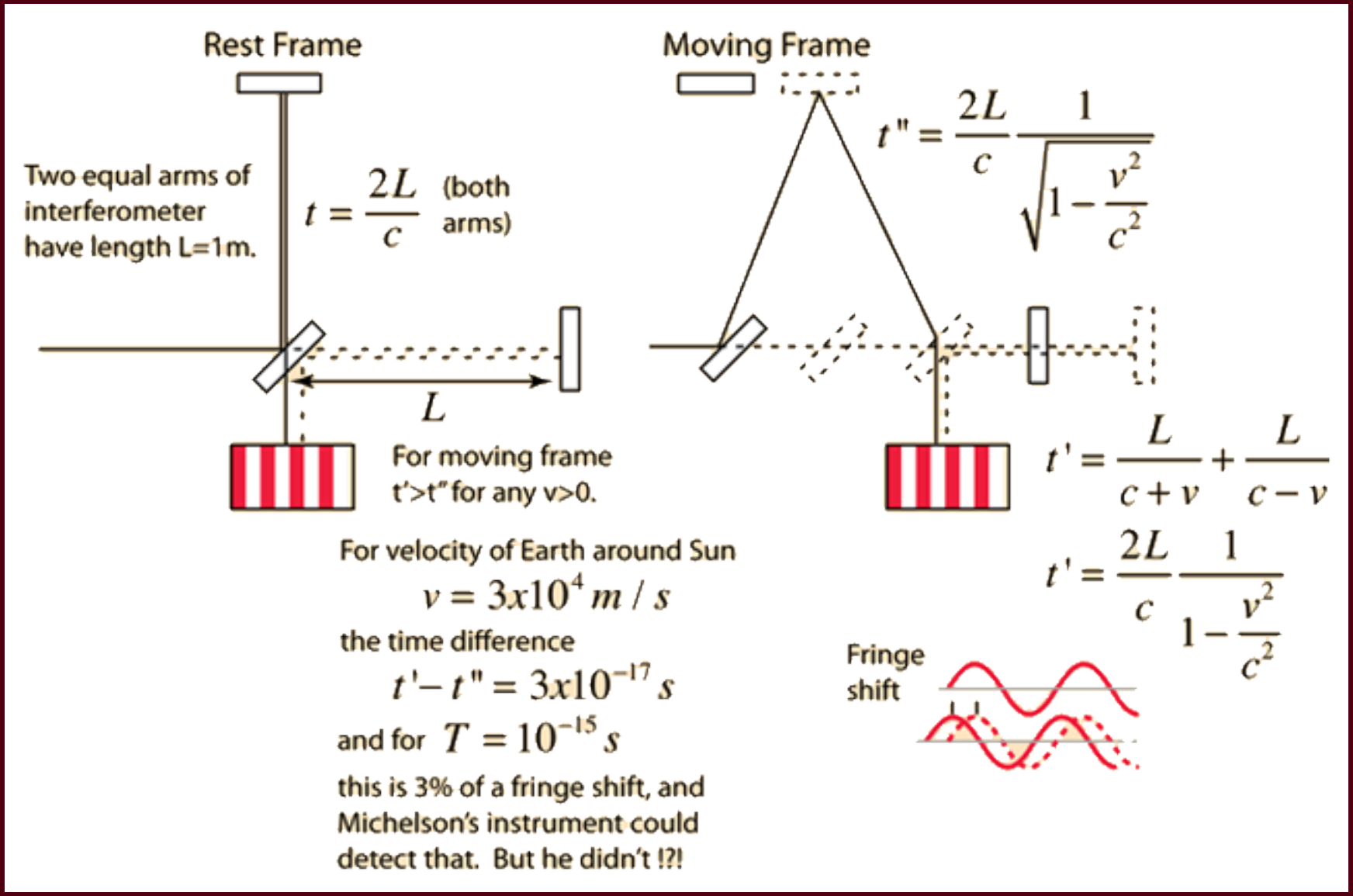
Maxwell's equations predict that light will travel with speed c, regardless of the reference frame (no 'ether')!

Michelson-Morely Experiment

- Attempted to measure the motion of the Earth relative to stationary luminiferous ether, the postulated medium for propagation of light.
- Light should travel slower or faster depending on its motion with or against the ether wind; think of fighting a headwind or enjoying a tailwind on your bicycle.
- By splitting a beam of light into two perpendicular paths and recombining them, a differential speed of light along each path would cause a phase shift, which could be detected by measuring the resulting interference pattern.
- Despite being repeated at different times of the day and over the course of years, when the ether wind was expected to change direction and magnitude, no phase shift was ever observed.



Michelson-Morley Experiment

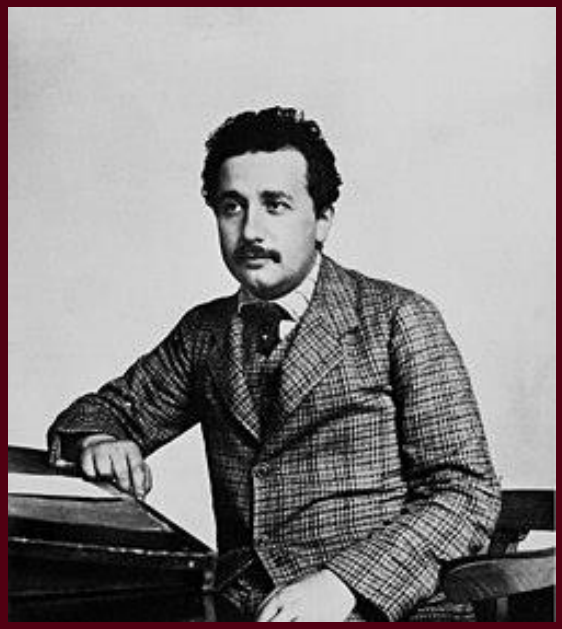


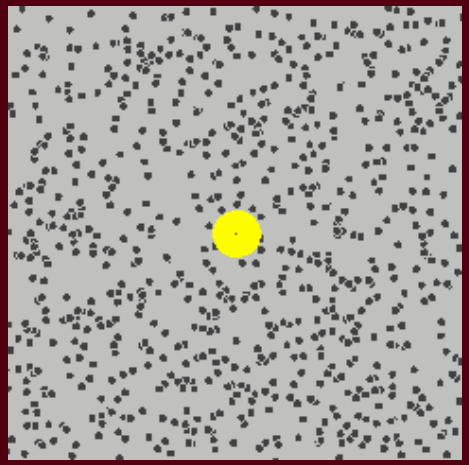
Albert Einstein (1879 – 1955)

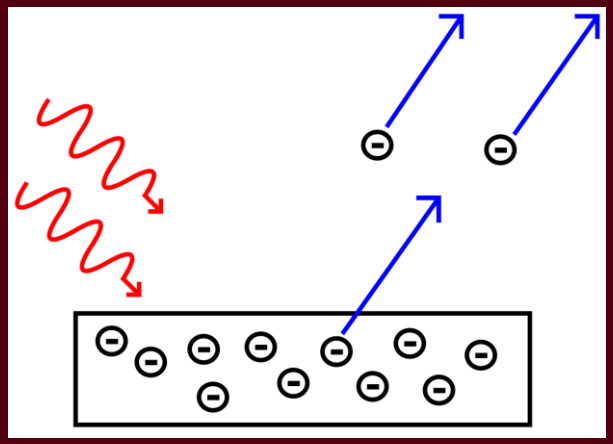
German physicist who, throughout his life, continually reshaped our understanding of space, time, matter, and energy.

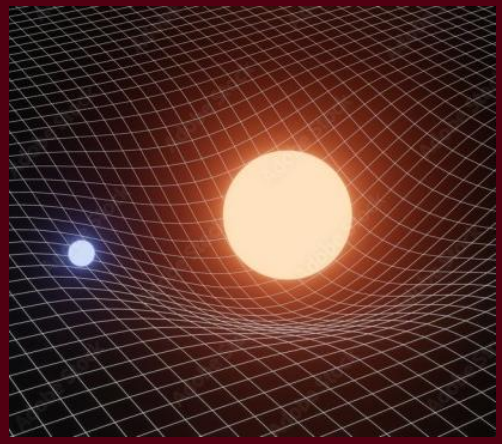
- Special relativity (1905)
- Brownian motion (1905)
- Photoelectric effect (1905)
- General relativity (1915)

If we can't detect the ether, and we can do physics without it, maybe it's just not there!

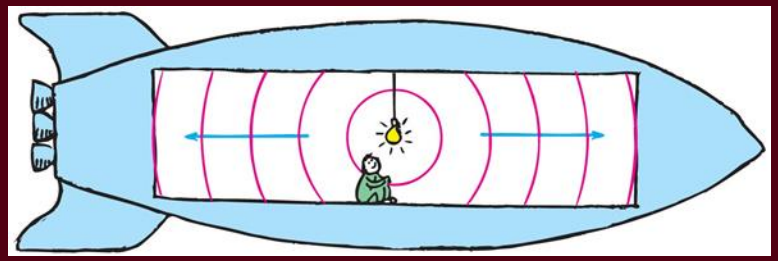


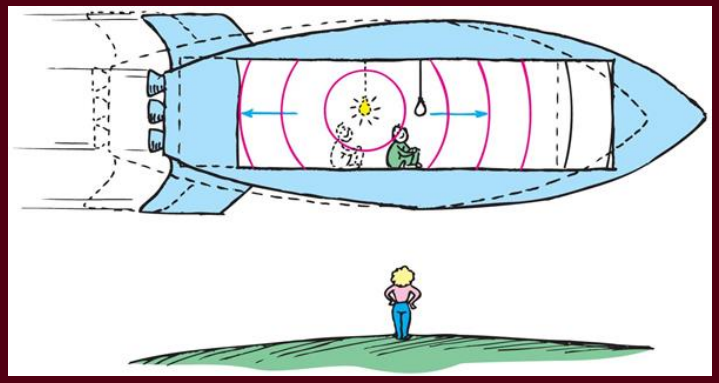






Relativity of Simultaneity





- From the point of view of the observer who travels with the compartment, light from the source travels equal distances to both ends of the compartment and therefore strikes both ends simultaneously.
- Because of the ship's motion, light that strikes the back of the compartment doesn't have as far to go and strikes sooner than light that strikes the front of the compartment.

Conceptual Question

Is the non-simultaneity of hearing thunder after seeing lightning similar to relativistic non-simultaneity?

A. Yes, it is exactly the same phenomenon.

B. No, it is a completely different phenomenon.

C. It depends upon how loud the thunder is.

D. It depends upon how far the thunder is.

Conceptual Question

Suppose that an observer standing on a planet sees a pair of lightning bolts simultaneously strike the front and rear ends of the compartment in a high-speed rocket ship. Will the lightning strikes be simultaneous to an observer in the middle of the compartment in the rocket ship?

A. Yes, they will be simultaneous.

B. No, they will be nonsimultaneous.

C. It depends upon how fast the ship is moving.

D. It depends upon how long the ship is.

Relativity of Simultaneity



Spacetime

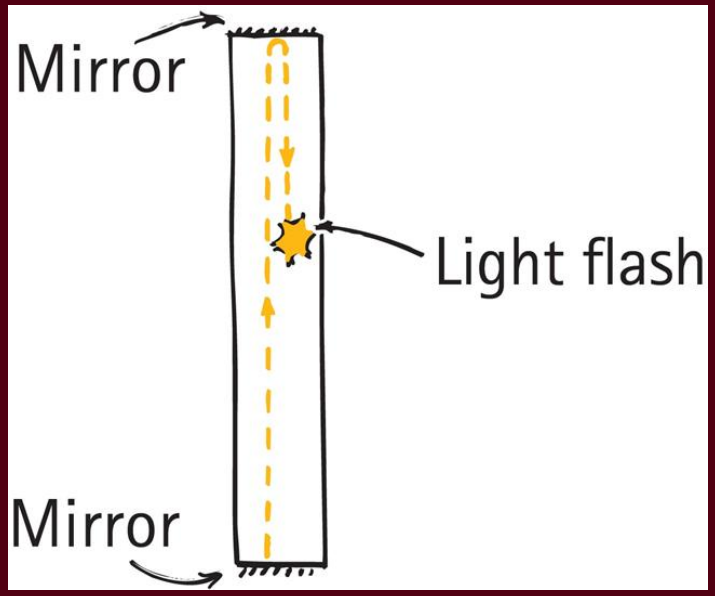
One observer's measurements of distance and time differ from similar measurements by another observer in a different reference frame.

Both measure the same ratio of the distance to the time separating events linked by a light beam: a greater measured distance in space corresponds with a greater measured time interval.

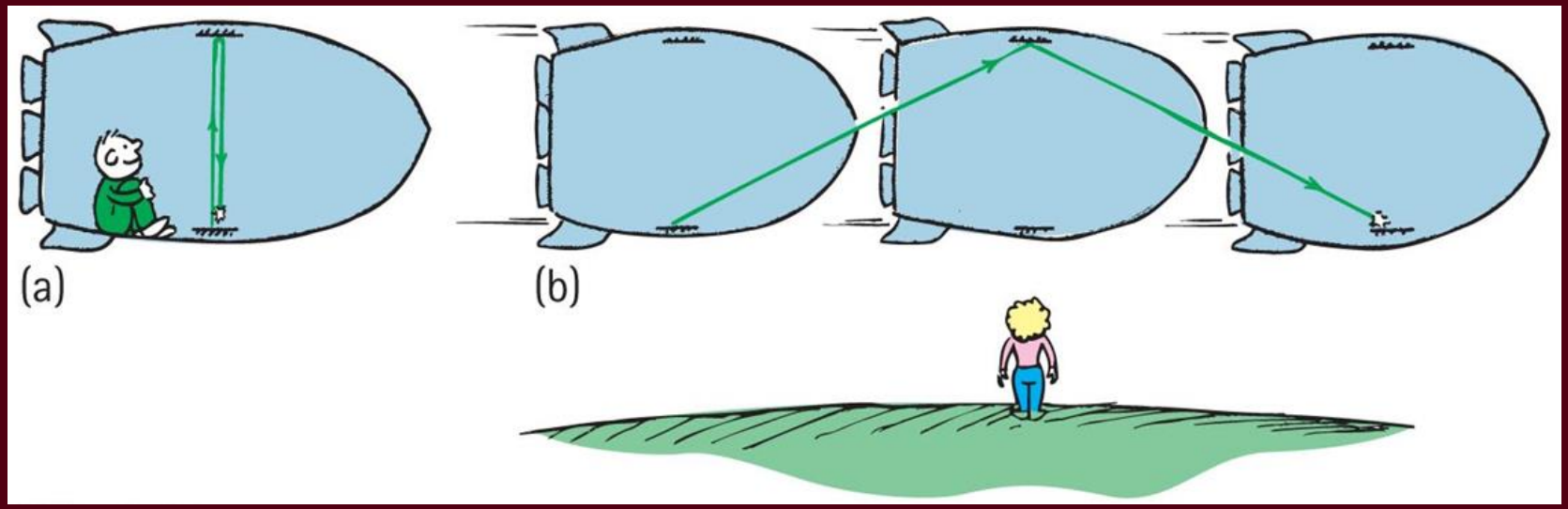
$$c = \frac{\text{distance}}{\text{time}} = \frac{\text{distance}}{\text{time}}$$

Light Clock

Imagine that we are somehow able to observe a flash of light bouncing up and down between a pair of parallel mirrors. If the distance between the mirrors is fixed, then the arrangement constitutes a light clock, because the back-and-forth trips of the flash take equal time intervals.



Time Dilation

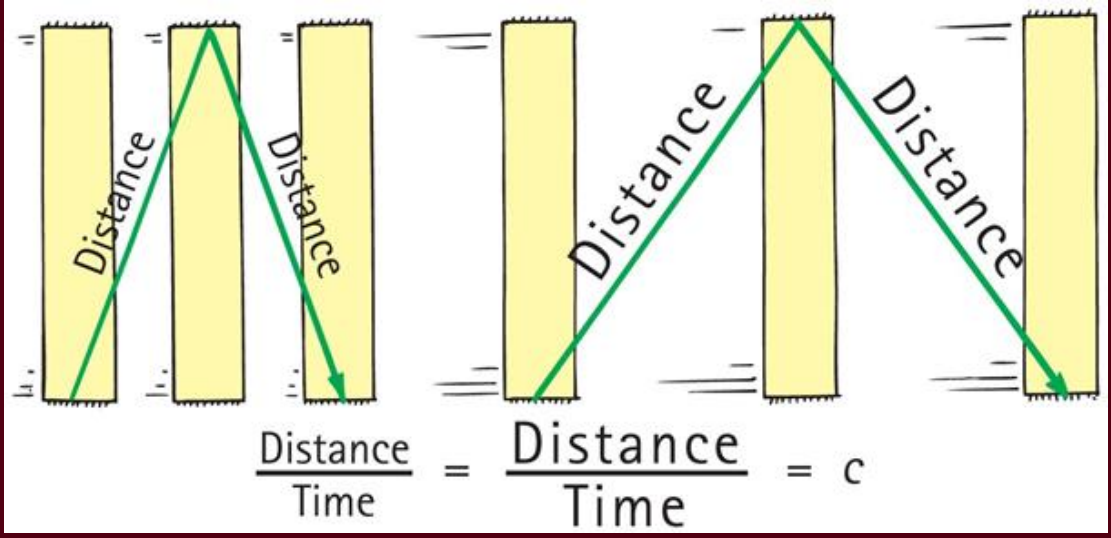


(a) An observer moving with the spaceship observes the light flash moving vertically between the mirrors of the light clock.
(b) An observer who sees the moving ship pass by observes the flash moving along a diagonal path.

Time Dilation

Observers agree that light traveled at speed c between two ticks of the clock, but they disagree about how far the light went, so they must disagree about how much time passed.

In fact, the observer who sees the clock moving will conclude that more time passed between two consecutive ticks of the clock.



Time Dilation Animation

Imagine a ball moves with speed c between two walls of a container, and does so independently of the motion of the container



The clock ‘ticks’ each time the ball hits the bottom wall

For every 5 ticks of the stationary clock, there are only 4 ticks of the moving clock!

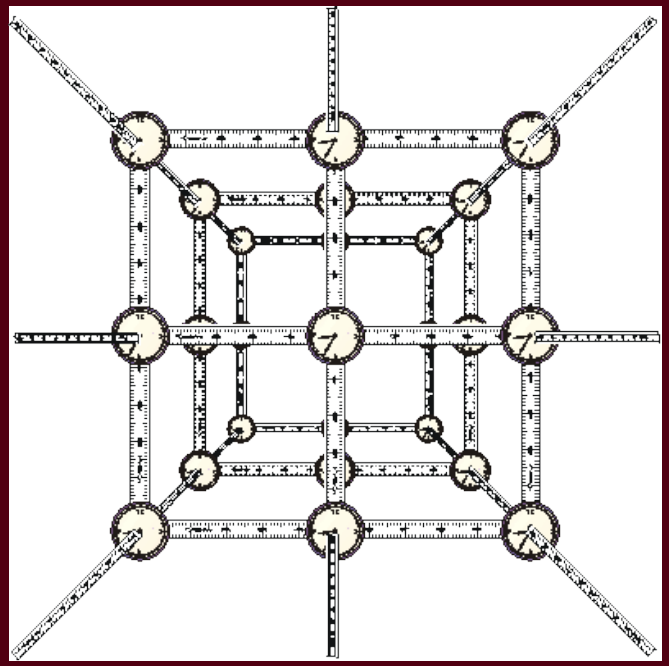
In Newtonian physics, the ball should not have the same speed in both frames, hence why the result is counterintuitive

Proper Time vs. Coordinate Time

Proper time is the time between two events that occur at the same location, as measured by a clock that is present at both events. It's like the time between when you wake up and go to sleep *as measured by the watch you carry*.

Coordinate time is the time measured between two events according to a system of *stationary synchronized clocks*.

- A proper time interval is always a coordinate time interval, but not all coordinate time intervals are proper time intervals (due to time dilation).
- Events may occur at different locations or the same location.



Conceptual Question

If you were moving in a spaceship at a high speed relative to Earth, would you notice a difference in your pulse rate or the pulse rate of people on Earth?

A. Yes, you would notice a difference in both pulse rates.

B. You would notice a difference in your pulse rate, but not the pulse rate of people on Earth.

C. You would notice a difference in the pulse rate of people on Earth, but not in your own pulse rate.

D. You would not notice a difference in either pulse rate.

Conceptual Question

Will observers A and B agree on measurements of time if observer A moves at half the speed of light relative to observer B?

A. Yes, they would agree completely.
B. No, they would disagree completely.
C. They would agree half of the time and disagree the other half of the time.
D. None of the above is correct.

Conceptual Question

Will observers A and B agree on measurements of time if both A and B move together at half the speed of light relative to Earth?

A. Yes, they would agree completely.

B. No, they would disagree completely.

C. They would agree half of the time and disagree the other half of the time.

D. None of the above is correct.

Conceptual Question

Does time dilation mean that time really passes more slowly in moving systems or only that it seems to pass more slowly?

A. Time really passes more slowly in moving systems.
B. Time only seems to pass more slowly in moving systems.
C. It depends upon how fast the system is moving.
D. It depends upon the direction in which the system is moving.

Conceptual Question

The Olympic Games is a two-week long sports competition. An interested alien astronomer watches the Olympics from a distant planet moving at high speed relative to Earth. If the alien were to compensate for the time the light from Earth takes to reach them, they would measure the length of the Olympics to be:

A. Greater than two weeks
B. Equal to two weeks
C. Less than two weeks

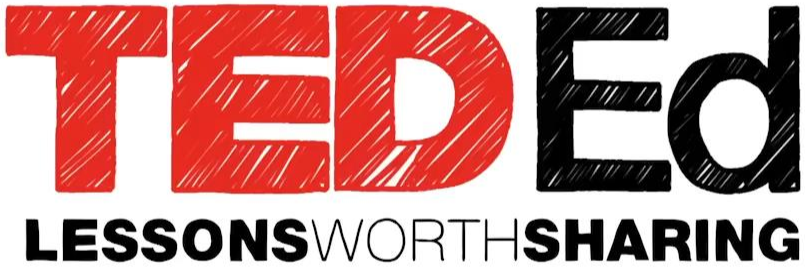
The Twin Paradox

- Identical twins, one an astronaut who takes a high-speed round-trip journey while the other stays home on Earth:
  - When the traveling twin returns, he's younger than the stay-at-home twin! How much younger depends on the relative speed.
- Since motion is relative, why doesn't the effect work equally well the other way around? Why wouldn't the traveling twin return to find his stay-at-home twin younger than himself?



Physics can be non-intuitive but never contradictory!

Resolution to the Twin Paradox



Conceptual Question

A ship sends equally spaced flashes every 6 minutes while approaching a receiver at constant speed. How will these flashes be spaced when they encounter the receiver?

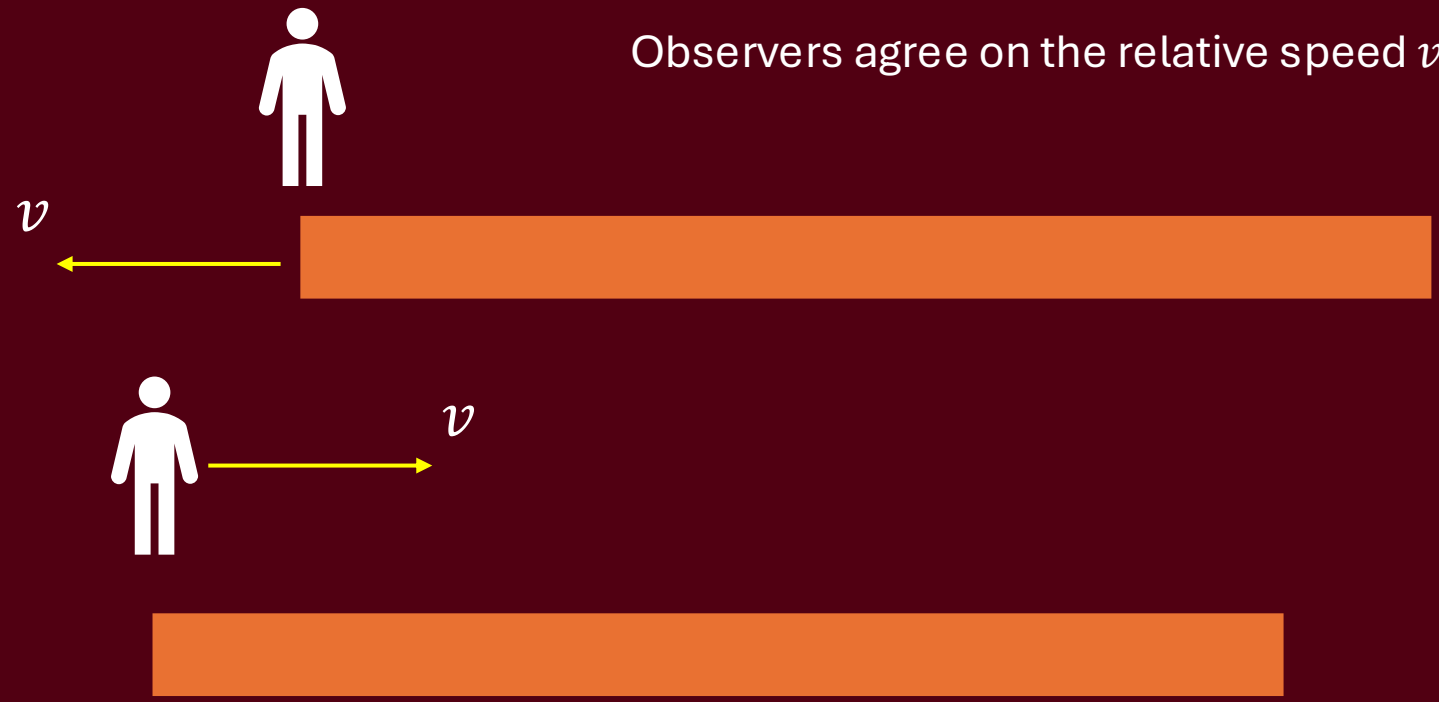
A. They will be equally spaced 6 minutes apart.
B. They will be equally spaced less than 6 minutes apart.
C. They will be equally spaced more than 6 minutes apart.
D. They will not be equally spaced.

Conceptual Question

Since motion is relative, can't we say as well that the spaceship is at rest and the Earth moves, in which case the twin on the spaceship ages more?

A. Yes.
B. No.
C. It depends on how fast the ship is moving.
D. It depends upon the direction in which the ship is moving.

Length Contraction



Observers agree on the relative speed v

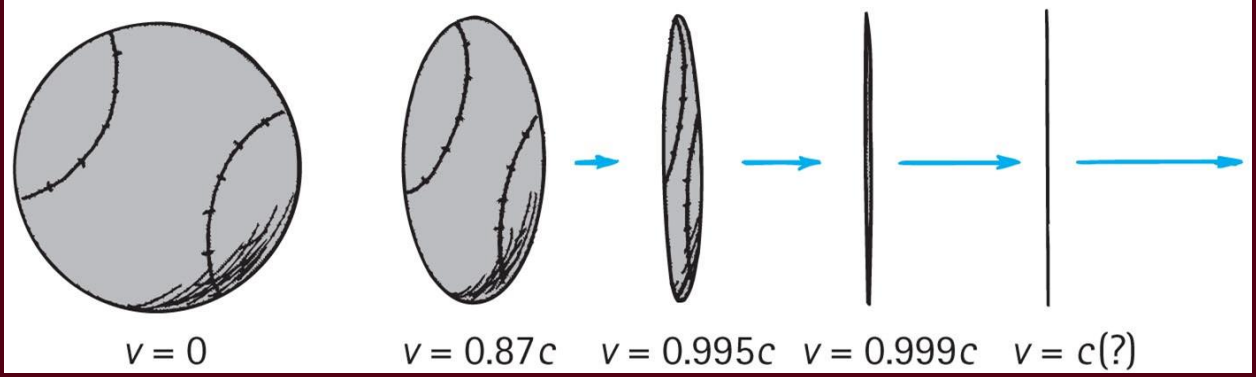
Bob's rest frame: both ends of the rod pass him in time interval $\Delta t_{\mathrm{Bob}} = L_{\mathrm{Bob}}/v$
Rod's rest frame: Bob passes both ends of the rod in time interval $\Delta t_{\mathrm{rod}} = L_{\mathrm{rod}}/v$
Due to time dilation, $\Delta t_{\mathrm{rod}} > \Delta t_{\mathrm{Bob}}$ (imagine Bob is holding a light clock)

$\therefore\ L_{\mathrm{rod}} > L_{\mathrm{Bob}}$

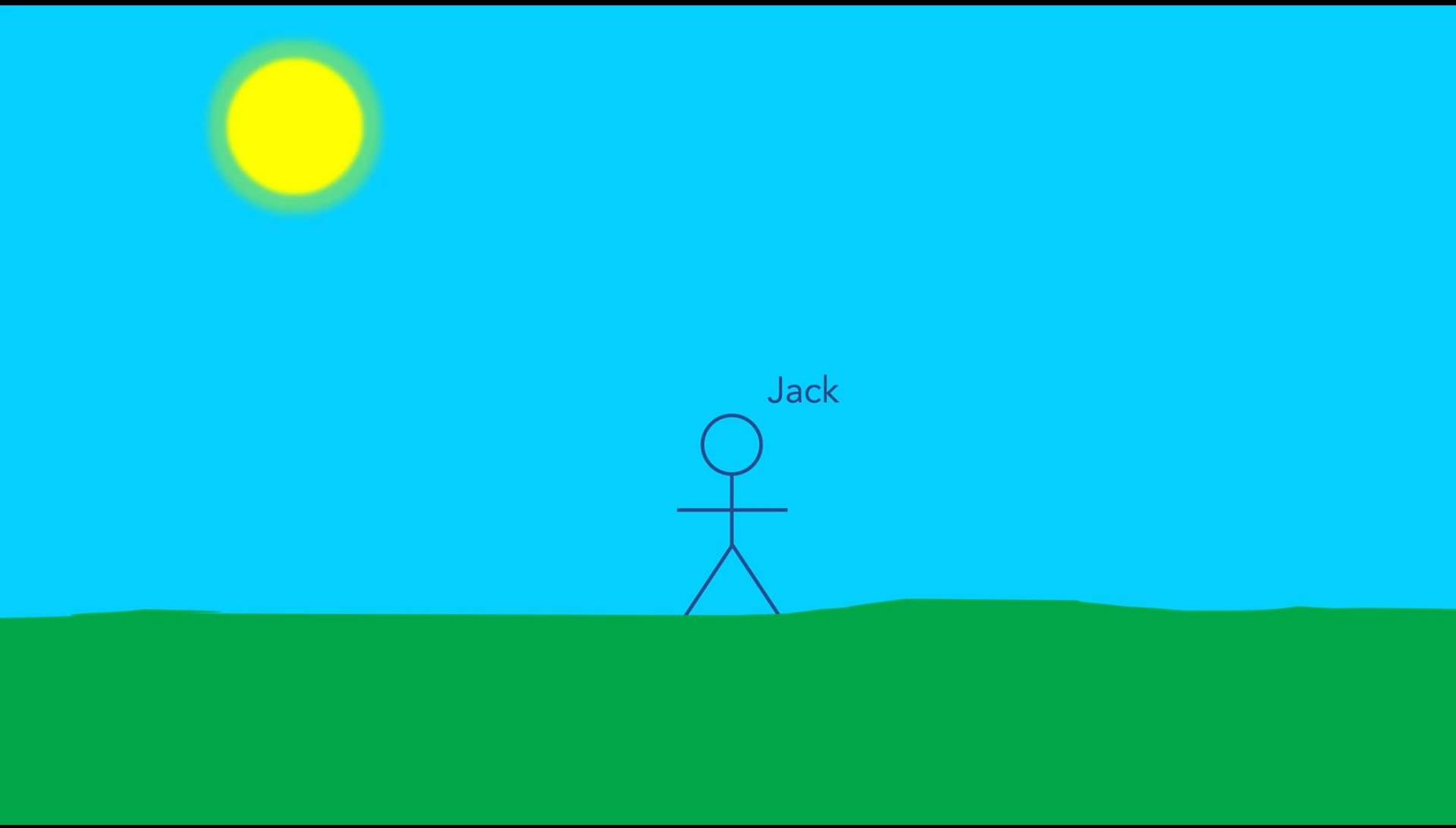
Bob measures the moving rod to have a *shorter length* than when it's at rest!

Length Contraction

- A moving object's length is measured to be shorter than its proper length, which is the length as measured in the object's own rest frame.
- Length contraction takes place only in the direction of travel. An object traveling horizontally experiences horizontal contraction but not vertical contraction (due to isotropy of space).



Einstein’s Ladder



Summary

Concept	Newtonian Physics	Special Relativity
Simultaneity	Events simultaneous in one frame are simultaneous in all frames. Time is absolute.	Observers in relative motion disagree about whether spatially separated events occur at the same time. There is no universal "now."
Time Dilation	All observers agree on elapsed time between events. Identical clocks always run at the same rate.	Moving clocks run slower relative to a stationary observer. Time intervals depend on relative motion.
Length Contraction	Object lengths are independent of observer motion.	Objects are shorter along the direction of motion when measured by a moving observer.
Space and Time	Separate, absolute entities.	Unified into *spacetime*.
Coordinate transformations	Galilean Transformations	Lorentz Transformations